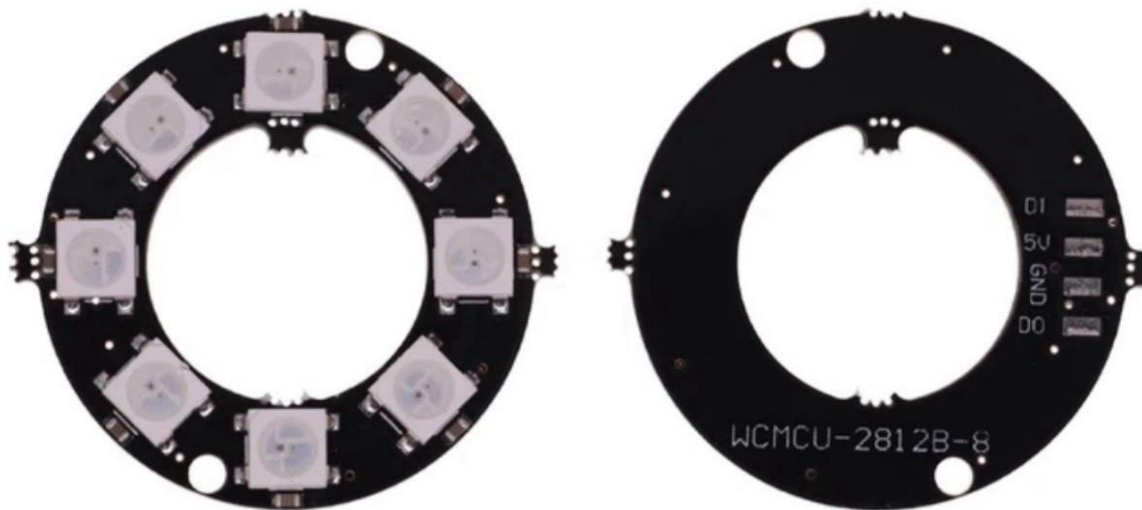


WS2812B RGB Neopixel LED Ring

These LED strips are just about the best way to get tons of colorful LED light with a minimum of wiring and fuss! Each ring is 3cm in diameter and contains 8 light elements. Each element contains a red, green, and blue LED, as well as a tiny WS2812 controller chip. This means that each element is individually addressable and can accept brightness values of 0-255 for each of its LEDs, giving you 24 bits of color precision overall. Each LED is activated by using a special serial data protocol. This protocol allows many LEDs to be daisy chained together so that one microcontroller can control the whole lot – with a single data signal. Any number of these LEDs can be chained together by connecting one module's data-out (DO) pin to another's data-in (DI) pin. This LED is compatible with Neopixel & FastLED Arduino Library.



Brief Data:

- Ring Diameter: Ø3cm.
- Ring of 8 pieces of 5050 Red, Green and Blue (RGB) LEDs.
- Each LED element is individually addressable with 24-bit color (8 bits each for R, G & B), 16 million colors.
- 3-Wires easy connection, 2-power supply lines and 1 data line.
- Interface: Serial (Shift Register).
- Compatible with Neopixel & FastLED Arduino Library.
- Data Speed: 800Kbps.
- Mounting Hole: 4-position M2 Screw.
- Power Supply: 5V.

Application Introducing to WS2812B LED Strip with Arduino:

This is the coolest type of LED strips in the market today. You can control the brightness and the color of each LED individually, which allows you to produce amazing and complex effects in a simple way. This LED strip is made by WS2812B LEDs wired in series. These LEDs have a IC built right into the LED. This allows a communication via a one-wire interface. This means that you can control lots of LEDs using just one digital pin of your Arduino. In the following figure you can see the chip inside the LED. The LED is an RGB LED and works like so.



Powering the WS2812B LED Strip:

The LED ring should be powered using a 5V power source. At 5V, each LED draws about 50mA, when set to its full brightness. This means that for every 8 LEDs, the strip may draw as much as 1.2 A. Make sure you select a power source that matches the module's needs. If you end up using an external supply, don't forget to connect the power supply ground to the Arduino ground.

Arduino Schematics:

In this example, the WS2812B LED ring will be powered using the 5V from Arduino board. In this case, we are controlling 8-LEDs per ring. Bear in mind that if you want to control many LEDs, you'll need to use an external power supply.

